

MICROLAND

Fundamentals of AI-Ops

4-Day Instructor-Led Online Training

Duration	Audience	Prerequisites
4 Days / 16 Hours	IT Service Professionals 1-5 Years Experience	No AI or programming knowledge required

Domains: Wintel | Networking | Security | Incident Management | Virtualisation

Aligned to: PeopleCert / DevOps Institute AIOps FoundationSM Certification Blueprint (All 8 Domains)

Revised Edition — Restructured from 5-Day / 20-Hour Programme

Programme Design Note: Why 4 Days?

This programme was originally designed as a 5-day, 20-hour course. A content analysis of the original structure found that Days 1 and 2 were both doing the same job — building conceptual groundwork — with no machine learning, no anomaly detection, and no automation content in either day. Keeping them separate created an imbalanced programme where participants spent two full days on theory before touching any AIOps-specific functionality.

The 4-day, 16-hour redesign merges those two conceptual days into a single Day 1, redistributes the content tightly, and protects all hands-on lab time. The result is a leaner programme that covers the same ground more efficiently.

What Changed	Original (5-Day)	Revised (4-Day)
Total scheduled time	20 hours	16 hours
Theory content	565 min (52%)	490 min (51%)
Hands-on lab time	360 min (33%)	310 min (32%)
Admin / buffer / Q&A	155 min (14%)	130 min (14%)
Blueprint domains covered	All 8	All 8 — nothing dropped
Lab notebooks	4 Colab notebooks	4 Colab notebooks — all retained
Capstone lab	90 min (Day 5)	90 min (Day 4) — fully preserved

❏ Important Note

Nothing substantive was removed in this restructure. All PeopleCert blueprint domains, all lab notebooks, the capstone, the final quiz, Generative AI content, DORA metrics, and the organisational change management module are fully preserved.

The reduction comes from consolidating two conceptual days into one and trimming admin buffer — not from cutting content.

About This Training Programme

The IT operations landscape is undergoing a fundamental shift. AI-powered platforms are replacing manual, reactive monitoring with intelligent, self-learning systems that detect anomalies before users feel them, correlate thousands of related alerts into a single root cause, and trigger automated fixes without human intervention.

This 4-day programme is designed for IT service professionals at Microland — Wintel engineers, network administrators, security analysts, incident managers, and virtualisation specialists — who need to understand, work within, and champion AIOps adoption. No prior AI or programming knowledge is assumed.

The curriculum is fully aligned with the eight domains of the PeopleCert / DevOps Institute AIOps FoundationSM certification, giving participants a direct path to an industry-recognised credential.

Programme at a Glance

Day	Theme	Blueprint Domains	Lab Notebook
Day 1	AIOps Foundations + Big Data	Domain 1: AIOps Foundation Domain 2: AIOps in the Org Domain 3: Big Data	day2_log_generator.ipynb (1,000 Windows log events)
Day 2	Machine Learning & Generative AI in AIOps	Domain 4: ML & Generative AI	day3_spike_simulator.ipynb (CPU spike, 720 data points)
Day 3	Use Cases, Incident Management & Org Mindset	Domain 5: Operations Metrics Domain 6: Use Cases & Mindset	day4_incident_simulator.ipynb (4-hr brute-force scenario)
Day 4	Evaluating Impact, Implementing AIOps & Capstone	Domain 7: Evaluating Impact Domain 8: Implementing AIOps	day5_memory_leak.ipynb (72-hr RAM leak capstone)

Daily Time Budget

Day	Total	Theory	Lab	Admin / Q&A	Lab Notebook
Day 1	4 hours	145 min	45 min	50 min	day2_log_generator.ipynb
Day 2	4 hours	100 min	60 min	40 min	day3_spike_simulator.ipynb
Day 3	4 hours	85 min	60 min	55 min	day4_incident_simulator.ipynb
Day 4	4 hours	65 min	90 min	45 min	day5_memory_leak.ipynb
Total	16 hours	395 min	255 min	190 min	4 notebooks

Free Tools Used Throughout the Programme

Tool	Purpose	Access
Elastic Cloud (14-day trial)	Log and metric ingestion, ML anomaly detection, dashboards	cloud.elastic.co — no credit card needed
Google Colab	All four lab data-generation notebooks — no local install needed	colab.research.google.com — free with Google account
Grafana Cloud (Free tier)	Metrics visualisation and alerting (reference tool)	grafana.com/auth/sign-up
webhook.site	Test webhook / automation triggers in the browser	webhook.site — free, instant
ServiceNow Developer Instance	ITSM ticketing and alert integration demo (Day 3)	developer.servicenow.com — free personal instance

Pre-Course Setup Checklist

- Create a free Elastic Cloud account at cloud.elastic.co (14-day trial, no credit card)
- Sign in to Google Colab at colab.research.google.com using any Google account
- Optional: Create a free Grafana Cloud account at grafana.com/auth/sign-up
- Optional: Create a ServiceNow Developer Instance at developer.servicenow.com
- Laptop with Chrome browser recommended — no corporate software installation required

❑ Important Note

All hands-on labs are browser-based. No VPN, no local installs, and no corporate IT approval is needed to participate.

The instructor will share all four Colab notebooks via a Google Drive link at the start of Day 1.

Day 1: AIOps Foundations + Big Data for IT Operations

Duration: 4 Hours | Format: Lecture + Demo + Hands-On Lab

Blueprint Domains: Domain 1 — AIOps Foundation | Domain 2 — AIOps in the Organisation | Domain 3 — Core Technologies: Big Data

Goal: Build the complete conceptual foundation for the entire programme. Understand where AIOps came from, what problems it solves, how it relates to DevOps and SRE, where your organisation sits on the maturity curve, and why big data principles are the prerequisite for any ML to work reliably.

❏ Important Note

Why are Foundations and Big Data covered together? Both topics are conceptual groundwork with no ML or automation content. Teaching them on the same day creates a single, coherent 'before the AI' context block, so Day 2 can focus entirely on machine learning without revisiting fundamentals.

Time	Module	Key Topics
0:00-0:30	Introduction + Programme Overview	Course objectives, tools setup check, how the 4 days connect, pre-course checklist review
0:30-1:00	Module 1.1	Evolution of IT Operations: Manual Monitoring > ITOA > AIOps; Gartner's definition; what changed in 2016
1:00-1:30	Module 1.2	Three Pillars (Observe, Think, Act); AIOps vs DevOps vs SRE vs MLOps; AIOps platform architecture
1:30-1:50	Module 1.3	Alert fatigue explained; the static-threshold problem; AIOps Maturity Model (Levels 0-4)
1:50-2:20	Module 2.1	The Five V's of Big Data applied to IT operations; why IT data is a big data problem
2:20-2:50	Module 2.2	Data sources: Metrics, Logs, Traces, Events; the Beats agent family; structured vs unstructured data
2:50-3:10	Module 2.3	Data normalisation; Elastic Common Schema (ECS); mapping Windows Event IDs and Cisco Syslogs
3:10-3:55	Lab	Elastic Cloud setup; Metricbeat configuration; run <code>day2_log_generator.ipynb</code> ; first Kibana dashboard
3:55-4:00	Wrap-up	Q&A; Day 2 preview; homework: explore your Kibana dashboard before tomorrow

Module 1.1 — The Evolution from Manual Monitoring to AIOps

To appreciate AIOps, you need to understand the journey that produced it. IT monitoring has evolved through three distinct generations, each driven by a fundamental change in the scale and complexity of IT environments:

Era	Approach	Key Limitation	What Broke It
Manual Monitoring (~1990s-2005)	Engineers log in manually and check dashboards; alerts sent by email or pager	Does not scale beyond a handful of servers; entirely dependent on human availability	Virtualisation created 10x more systems to monitor overnight
IT Operations Analytics (ITOA) (~2005-2016)	Centralised log and metric collection; static threshold-based rules and alerts	Can only analyse past data with fixed rules; cannot adapt to dynamic environments	Cloud and containers created environments that change topology every minute
AIOps (2016-present)	Machine learning learns normal behaviour dynamically; correlates signals across all sources; enables prediction and automation	Requires good data quality and cultural readiness to deliver full value	Still evolving — Generative AI is now extending AIOps capabilities further

Gartner first introduced the term in 2016 as 'Algorithmic IT Operations', emphasising the shift from human-written rules to algorithm-driven analysis. Within a year the term was refined to 'Artificial Intelligence for IT Operations', reflecting the growing role of machine learning.

Gartner's formal definition: An AIOps platform combines big data and machine learning functionality to support all primary IT operations functions through the scalable ingestion and analysis of the ever-increasing volume, variety and velocity of data generated by IT.

□ Key Takeaway

AIOps is not a single product you purchase — it is an operational approach that combines big data infrastructure, machine learning models, and automation capabilities.

ITOA failed because it applied static rules to historical data. AIOps succeeds because it learns continuously from live data and updates its model of 'normal' as the environment changes.

Module 1.2 — Core Concepts and the AIOps Ecosystem

The Three Pillars: Observe, Think, Act

Every concept in this programme maps back to one of three pillars. Keep these in mind as the anchor framework throughout all four days:

- **OBSERVE:** Ingest all operational signals — metrics, logs, traces, events — from every layer of the IT stack into a unified platform. The quality of what you observe determines the quality of every downstream outcome.
- **THINK:** Apply machine learning to understand patterns, detect anomalies, correlate related events, identify probable root causes, and predict future problems — without human-defined rules.
- **ACT:** Trigger appropriate responses automatically — a notification, an ITSM ticket, a remediation playbook, or a self-healing script — based on what the ML has identified.

AIOps and Its Neighbours — The Ecosystem Explained

Term	Focus	Relationship to AIOps
DevOps	Culture and engineering practices for faster software delivery; CI/CD pipelines	AIOps observes and improves the operational side of DevOps pipelines; they are complementary, not competing
SRE (Site Reliability Engineering)	Engineering discipline applying software practices to IT operations; error budgets and SLOs	AIOps provides the intelligent data layer that SREs use to make reliability decisions at scale
MLOps	Managing the lifecycle of ML models in production: training, deployment, monitoring, retraining	AIOps consumes ML models. MLOps manages those models. Think of MLOps as the engine room for AIOps brains.
AIOps	Using AI and ML to enhance IT operations: monitoring, incident management, automation	The operational umbrella where all three above converge in a production IT environment

□ Key Takeaway

AIOps does not replace DevOps or SRE — it amplifies them by providing intelligent, real-time data analysis that those disciplines need to function at scale.

Understanding the AIOps-MLOps relationship is important: when your anomaly detection model becomes inaccurate (for example after a major infrastructure change), it is an MLOps problem — the model needs retraining.

Module 1.3 — The Problem AIOps Solves and the Maturity Model

Alert Fatigue: The Core Problem

If you work in Wintel, network, or security operations at Microland, you almost certainly experience this daily: a flood of monitoring alerts, most of which are noise, obscuring the handful that matter. A 2023 industry study found the average enterprise IT team receives over 2,500 alerts per day, of which only 20-30% require any action. Engineers spend up to 67% of their time triaging alerts that turn out to be irrelevant. The root cause is the static threshold — a fixed rule like 'if CPU > 90%, alert' that treats every occurrence identically, regardless of context, time of day, or workload pattern.

The AIOps Maturity Model — Where Are You Today?

Level	Stage	Characteristics	Typical IT Operations Reality
0	Reactive	Alert storms, manual triage, no correlation. Each team has its own siloed tools.	Individual monitoring dashboards per team; no central view; P1 detected by end users before engineers
1	Detect	Centralised data collection; basic event correlation; first ML anomaly detection jobs running.	Elastic or Splunk centralising logs; first ML jobs detecting obvious spikes; alert volume starting to fall
2	Proactive	Predictive insights from ML; forecasting used for capacity planning; root cause surfaced automatically.	ML flagging memory leaks before service crashes; disk exhaustion raised as proactive change requests

Level	Stage	Characteristics	Typical IT Operations Reality
3	Preventive	Automation triggered by AI insights; approved self-healing playbooks; MTTR under 5 minutes for infrastructure failures.	Auto-restart playbooks for approved low-risk services; teams consistently trust ML recommendations
4	Autonomous	Systems self-optimize; AI makes and executes most routine operational decisions independently.	Fully autonomous remediation with human oversight only for non-standard or high-risk situations

❏ Important Note

Most large IT service organisations today sit between Level 0 and Level 1. The realistic goal for this programme is to equip you to work confidently at Level 1-2 and understand the roadmap to Level 3.

Reaching Level 4 is a multi-year journey that requires not just technology but significant cultural and process change. Expecting to jump from Level 0 to Level 4 after a tooling purchase is the most common AIOps failure pattern.

Module 2.1 — Big Data: Why IT Operational Data Is a Different Problem

The term 'big data' is associated with customer analytics and social media — but your IT environment generates big data every second of every day. To understand why this matters for AIOps, consider the Five V's framework and apply each dimension to your own operational environment:

The V	Definition	IT Operations Example	AIOps Implication
Volume	The sheer quantity of data generated	A 1,000-server estate generates approximately 500 million log events per day	No human can read this at scale; ML must process it — human triage is the bottleneck, not the solution
Velocity	The speed at which data arrives and must be processed	Network flows, SNMP traps, and Windows events arrive in real time, continuously	AIOps must ingest and analyse data with sub-second latency to enable real-time anomaly detection
Variety	The diversity of data types and formats from different sources	Windows Event XML, Cisco Syslog plain text, VMware JSON, SNMP binary, CSV exports from legacy tools	All formats must be normalised into a common schema before ML can correlate across them
Veracity	The accuracy and trustworthiness of the data	A misconfigured NTP clock causes all timestamps from one server to be 4 hours off, corrupting ML model training	Data quality is foundational — poor veracity degrades ML accuracy more than any other single factor
Value	The actionable insight extractable from high-volume data	From 500 million events, only ~200 require human attention — if ML can reliably identify them	The entire purpose of AIOps is to extract that 200 from the 500 million efficiently and accurately

The V	Definition	IT Operations Example	AIOps Implication
❏ Key Takeaway The Five V's explain precisely why traditional monitoring tools fail at scale: they were designed for low-volume, low-variety, manually-reviewed data. AIOps platforms are designed for all five V's simultaneously. Veracity is the most underappreciated V in practice. Before enabling ML, audit your data quality: missing fields, wrong timestamps, and inconsistent naming degrade ML accuracy more than any algorithm limitation.			

Module 2.2 — Data Sources: What AIOps Observes

AIOps consumes three primary signal types, commonly called the Observability Triangle, plus two additional operational signals:

- **Metrics:** Numeric, time-stamped measurements sampled at regular intervals. Fast to process, ideal for real-time baselines and forecasting. Examples: CPU %, network throughput Mbps, memory used GB, disk I/O operations per second.
- **Logs:** Timestamped text records of discrete events and state changes. Rich in context but unstructured and variable in format across sources. Examples: Windows Event Log, Cisco Syslog, application error logs, web server access logs.
- **Traces:** End-to-end records of a request's journey through distributed systems. Most relevant for application and microservices operations teams. Examples: HTTP request chains, database query execution paths, API call sequences.
- **Events:** Discrete state-change notifications — a VM power-off, a configuration change, a deployment completion. Provide organisational context alongside technical signals.
- **Tickets:** ITSM incident and change records from ServiceNow or Jira. Provide historical resolution context that ML can use to suggest remediation steps for similar future incidents.

Common Data Collection Agents — All Free and Open Source

Agent / Tool	What It Collects	Primary Audience at Microland
Metricbeat (Elastic)	CPU, RAM, disk, network interface metrics from any system	Wintel and infrastructure teams — live system health
Filebeat (Elastic)	Any log file from any path on disk — application, IIS, custom	All teams using file-based logging
Winlogbeat (Elastic)	Windows Event Log: Security, System, Application channels	Wintel and security teams
Packetbeat (Elastic)	Network packet data: DNS, HTTP, flow statistics	Networking teams — application-layer visibility
Prometheus Node Exporter	CPU, memory, filesystem, network metrics via HTTP scrape endpoint	Linux and cloud infrastructure teams

Agent / Tool	What It Collects	Primary Audience at Microland
OpenTelemetry Collector	Vendor-neutral collection of metrics, logs, and traces	Teams standardising on platform-agnostic observability

Module 2.3 — Data Normalisation: The Prerequisite for ML

This is the single most important data concept in the programme. Machine learning cannot work across heterogeneous data sources unless all data speaks a common language.

Consider: your Windows server logs a failed login as Event ID 4625 with a field called SubjectUserName. Your Cisco router logs the same type of event as %AAA-3-BADAUTH with the username inside the message body. Your Linux server logs it as 'sshd: Failed password for jdoe'. Three sources, three completely different formats, all describing the same security event. For ML to correlate these into a single 'failed login pattern' across your estate, they must be normalised into a common schema before indexing.

Elastic Common Schema (ECS) — The Normalisation Standard

Raw Source	Raw Field / Format	ECS Normalised Field
Windows Event 4625	SubjectUserName = jdoe	user.name = jdoe
Windows Event 4625	IpAddress = 10.10.1.5	source.ip = 10.10.1.5
Cisco AAA syslog	Parsed from message text	event.action = authentication_failure
Linux sshd log	Parsed from message text	event.outcome = failure

Key Windows Event IDs — AIOps Context

Event ID	Description	AIOps Use Case
4624	Successful logon	Baseline logon patterns; unusual time-of-day logons
4625	Failed logon attempt	Brute force detection — anomalous spike in failure count from one IP
4648	Logon with explicit credentials	Lateral movement indicator — unusual credential use pattern
7045	New service installed	Persistence detection — new service created at 3 AM is anomalous
4688	New process created	Behavioural anomaly — unexpected process on a production server
41	Kernel power unexpected shutdown	Cascade correlation — upstream cause of downstream VM unavailability

□ Key Takeaway

Data normalisation is not a one-time activity — it is an ongoing operational discipline. Every new data source onboarded to AIOps requires a normalisation mapping before ML results will be reliable for that source.

Event ID	Description	AIOps Use Case
	The quality of your AIOps outcomes is directly proportional to the consistency and completeness of your normalised data. Invest in data quality before investing in more ML jobs.	
	□ Hands-On Lab: Day 1 — Elastic Cloud Setup + First Data + Log Generator Duration: 45 minutes Platform: Elastic Cloud + Google Colab Objective: Get data flowing into Elastic Cloud and build your first Kibana dashboard. Part A — Elastic Cloud Setup (15 min): 1. Go to cloud.elastic.co and create a free trial deployment (choose nearest region). 2. Download Metricbeat from elastic.co/downloads — configure <code>cloud.id</code> and <code>cloud.auth</code>. 3. Run: <code>./metricbeat -e</code> (or <code>metricbeat.exe -e</code> on Windows). 4. In Kibana Discover: verify CPU metric events are flowing under the <code>metricbeat-*</code> index. Part B — Run the Log Generator Notebook (20 min): 1. Open <code>day2_log_generator.ipynb</code> from the instructor's shared Google Drive link. 2. Paste your Cloud ID and API Key into the configuration cell. 3. Run all cells in order. The notebook generates 1,000 Windows-style log events (including 20 injected brute-force anomalies) and uploads them to Elastic Cloud. 4. Verify in Kibana: index <code>aiops-lab-day2</code> should show 1,000 documents. Part C — First Dashboard (10 min): 1. Kibana > Dashboards > Create dashboard > Add visualisation. 2. Create a bar chart of <code>event.action</code> values from the <code>aiops-lab-day2</code> index. 3. Create a line chart of event count over time (last 4 hours). 4. Save the dashboard as 'Day 1 — Infrastructure Overview'. Challenge: The 20 anomalous brute-force events are hidden in 980 normal events. Try to find them using KQL: <code>source.ip : "185.220.101.47"</code> Can you see them clearly in your dashboard? (Day 2 ML will find them automatically.) Expected Outcome: Elastic Cloud running, two datasets ingested, first dashboard built.	
	? Check Your Understanding 1. Explain in one sentence why ITOA failed to handle cloud and container environments, and what fundamental capability AIOps added to address this. 1. A colleague says 'AIOps will replace DevOps in our organisation.' Is this accurate? Explain the correct relationship between AIOps, DevOps, and SRE. 1. Your Wintel team monitors 500 servers. Place your current environment on the AIOps Maturity Model (0-4) and describe one concrete action that would move you to the next level. 1. Explain the Five V's of big data using examples from your own daily work. Which V do you think is the hardest to manage in practice, and why?	

Event ID	Description	AIOps Use Case
	1. A Cisco router generates: '%SEC-6-IPACCESSLOGP: list BLOCK denied tcp 10.10.1.5(54321) -> 192.168.1.100(80)'. Map the key fields to their ECS equivalents.	

Day 2: Machine Learning and Generative AI in AIOps

Duration: 4 Hours | Format: Lecture + Demo + Hands-On Lab

Blueprint Domain: Domain 4 — Core Technologies: Machine Learning and Generative AI

Goal: Understand how machine learning enables the 'Think' pillar — without writing a single algorithm. Learn the critical distinction between deterministic and probabilistic approaches, interpret anomaly scores correctly, and understand how Generative AI is extending AIOps capabilities in 2025.

Time	Module	Key Topics
0:00-0:40	Module 3.1	ML for operations teams: supervised vs unsupervised; deterministic vs probabilistic; how models train
0:40-1:15	Module 3.2	Anomaly detection in practice: Elastic ML functions, anomaly scoring, the 'surprise meter' model
1:15-1:45	Module 3.3	Forecasting for capacity planning; time-series projection; disk exhaustion and memory trend examples
1:45-2:10	Module 3.4	Generative AI in AIOps: incident summarisation, natural language querying, runbook generation, guardrails
2:10-3:10	Lab	day3_spike_simulator.ipynb: create ML job, inject spike, interpret anomaly score, run 7-day forecast
3:10-3:40	Discussion	Deterministic vs probabilistic debate; when would you trust the ML? When would you override it?
3:40-4:00	Wrap-up	Q&A; Day 3 preview; homework: note one 'real' anomaly from your actual environment

Module 3.1 — Machine Learning Concepts for Operations Teams

You do not need to become a data scientist to work in an AIOps environment. You need to understand what the ML is doing — well enough to trust its outputs, interpret its scores, and know when to question or override it.

Supervised vs Unsupervised Learning

Type	How It Works	AIOps Application
Supervised Learning	You label training data as 'normal' or 'abnormal'. The model learns the difference from your labelled examples.	Used in spam filtering and fraud detection. Rarely the right choice in AIOps because you cannot label every possible failure type in advance — the environment is too complex.
Unsupervised Learning	No labels required. The model learns the statistical pattern of 'normal' from your historical data alone. Any significant deviation from that pattern is flagged automatically.	The primary ML approach in AIOps anomaly detection. The model learns what 'normal CPU on a Monday morning' looks like for your specific environment — without you defining it.

Deterministic vs Probabilistic Approaches — A Critical Distinction

This distinction is one of the most important concepts in the entire programme — and one that regularly causes confusion in post-incident reviews.

Approach	How It Works	When to Use It
Deterministic	Rule-based. The same input always produces the same output. 'If CPU > 90% for 5 consecutive minutes, raise a P2 alert.' Fully transparent, auditable, and reproducible.	Compliance-critical checks, known failure signatures with regulatory implications, and any automated action that must be fully explainable to an auditor.
Probabilistic (ML-driven)	Statistical. The model weighs many signals simultaneously and outputs a probability score. The same data point may score differently as the model continues learning from new data.	Detecting novel, unexpected patterns that you cannot anticipate or write rules for in advance. The primary mode for AIOps anomaly detection across complex, multi-dimensional environments.

A mature AIOps platform uses both approaches: deterministic rules for known, well-defined conditions that must always trigger; probabilistic ML for detecting the unexpected. Understanding which produced a given alert is essential for explaining it in a post-incident review.

□ Key Takeaway

Probabilistic ML does not give you certainty — it gives you probability scores. A score of 90 means 'this is highly statistically unusual given learned behaviour', not 'this is definitely a problem'. Always correlate ML outputs with other signals before taking automated action. When a senior engineer asks 'why did the AI alert on this?' — you must be able to answer. If the alert was probabilistic, the answer is: 'The ML flagged this because the value at this time was significantly outside the pattern it has learned as normal for this metric.'

Module 3.2 — Anomaly Detection in Practice

How Elastic ML Learns Your Baseline

- You specify: a data field (e.g. system.cpu.total.pct), a detection function (e.g. high_mean), and a time bucket (e.g. 15 minutes)
- The ML job ingests historical data and learns: what is the typical CPU value at 9 AM on a weekday? At 3 AM on a Sunday? What is the normal range of variation? What are the weekly and daily seasonal patterns?
- Seasonality is captured automatically — the model knows Monday mornings look different from Saturday nights without you configuring this
- After 2-4 weeks of data the model is reliable. Basic detection works sooner, but early false positive rates are higher

The Anomaly Score: Your 'Surprise Meter'

The anomaly score (0 to 100) measures how statistically surprising a data point is, given everything the model has learned about that metric. Think of it as: 'How surprised should the system be to see this value at this time of day on this day of the week?'

Score Range	Interpretation	Recommended Response
0 - 24	Normal — within expected range	No action needed
25 - 49	Slight deviation — monitor	Review if combined with other correlated signals
50 - 74	Notable anomaly — investigate	Correlate with logs and other metrics; open investigation
75 - 100	High anomaly — statistically very unusual	Escalate or trigger alert; correlate multiple signals for root cause before acting

❏ Important Note

A high anomaly score is NOT the same as a confirmed incident. A scheduled batch job running at an unexpected time, a capacity upgrade, or a bank holiday can all produce high scores without indicating a problem.

Context from correlated signals confirms whether an anomaly is genuinely problematic. Never escalate based on an anomaly score alone.

Key Detection Functions in Elastic ML

Function	What It Detects	Operational Example
high_mean	Value is higher than expected for this time period	CPU spike on a server that normally idles at weekends
low_mean	Value is lower than expected	Request rate drop — possible upstream dependency failure
high_count	More events than usual in the time window	Sudden spike in authentication failures — potential brute force
rare	A value that almost never appears in the data	A new process name executing on a production server
time_of_week	An event occurring at an unusual time for this day and hour	Admin account login at 3 AM on a Sunday morning

Module 3.3 — Forecasting for Capacity Planning

Beyond detecting current anomalies, Elastic ML can project a metric's future trajectory based on its historical pattern. For IT operations teams, this is the clearest path from reactive to proactive operations:

- Disk exhaustion: 'At current growth rate, this volume will reach 90% in 14 days' — raise a proactive change request today, not after the disk fills and a service crashes
- Memory pressure: 'RAM utilisation on this server has been growing 3% per week for the last month' — schedule a memory upgrade before performance degrades
- Network saturation: 'This WAN link will reach 80% utilisation during next month's quarter-end batch processing window' — plan bandwidth provisioning before the event
- Service capacity: 'Current application instance count will be insufficient during the projected peak load in 3 weeks' — scale proactively rather than reactively

□ Key Takeaway

Forecasting does not predict the future with certainty — it extrapolates from learned patterns. The confidence interval (the shaded band around the forecast line) shows the range of uncertainty. A wide band means the metric is highly variable and the forecast is less reliable.

Forecasting is only as good as the baseline. A model trained on 2 weeks of data will produce less reliable forecasts than one trained on 3 months of seasonally-complete data.

Module 3.4 — Generative AI in AIOps (2025 Context)

The integration of Large Language Models (LLMs) into AIOps platforms is the most significant development in the field in the past two years. These are the most practically useful Gen AI capabilities for Microland's operations teams today:

Capability	What It Does	Practical Value for Your Role
Incident Summarisation	Automatically generates a plain-English summary of an incident: what happened, which systems were affected, the event timeline, and the probable cause	Eliminates the 30 minutes an L2 engineer spends writing an initial incident summary from scratch; every incident arrives pre-summarised with full context
Natural Language Log Query	Ask questions in plain English: 'Show me all failed logins from external IPs in the last 6 hours' — instead of writing Elasticsearch Query Language	Lowers the skill barrier for operational investigation; L1 engineers can explore data without learning query syntax
Runbook Suggestion	Given an anomaly type and historical resolution patterns, Gen AI drafts a suggested runbook or step-by-step remediation guide	Accelerates response for novel incident types; reduces dependency on tribal knowledge held by senior engineers
Alert Explanation	Converts a cryptic ML anomaly alert into a human-readable explanation with suggested investigation steps	Helps L1 engineers understand what the ML found without needing to interpret raw probability scores

□ Important Note

Generative AI outputs in AIOps are probabilistic suggestions — not verified facts. Every Gen AI-produced incident summary, root cause statement, or runbook must be reviewed by a human before acting on it.

Data privacy is critical: before connecting a Gen AI assistant to your operational data, verify that your organisation's data governance policy permits it. Log data can contain usernames, IP addresses, session tokens, and other sensitive information.

□ Hands-On Lab: Day 2 — ML Anomaly Detection Job + Spike Analysis

Duration: 60 minutes | Platform: Elastic Cloud + Google Colab

Notebook: day3_spike_simulator.ipynb

Objective: Generate realistic CPU time-series data with an injected spike, upload it to Elastic, create an ML job, and interpret the anomaly score.

Capability	What It Does	Practical Value for Your Role
Part A — Generate and Upload the Data (15 min): 1. Open <code>day3_spike_simulator.ipynb</code> in Google Colab. 2. Paste your Cloud ID and API Key into the configuration cell. 3. Review the parameters: 6 hours of data, 30s intervals, 3x spike at the 3-hour mark. 4. Run all cells. The notebook generates 720 CPU metric samples and uploads them. 5. Observe the matplotlib chart — the spike should be visually obvious. Part B — Create the ML Anomaly Detection Job (25 min): 1. Kibana > Machine Learning > Anomaly Detection > Create Job > Single Metric. 2. Index: <code>aiops-lab-day3</code> Function: <code>high_mean</code> Field: <code>system.cpu.total.pct</code> 3. Bucket span: 15m Job ID: <code>day2-cpu-spike-detection</code> 4. Set time range to Full (all 6 hours). Click Create Job. 5. Open Anomaly Explorer. Locate the red/orange anomaly block at the 3-hour mark. 6. Click the anomaly block and record: the anomaly score, the actual value, the typical value. Part C — Create a 7-Day Forecast (10 min): 1. Go to Machine Learning > Single Metric Viewer > select your job. 2. Click Forecast > Duration: 7 days. 3. Observe the projected trend line and the confidence interval band. 4. Question: what does the width of the confidence interval tell you about data reliability? Optional Extension: Run the experiment cell at the end of the notebook with <code>NEW_MULTIPLIER = 1.5</code>. Create a second ML job on the new index. Compare: does a 1.5x spike still get detected? What is the anomaly score difference between a 1.5x and 3x spike? Discussion (10 min): - Is the alert from your ML job deterministic or probabilistic? How do you know? - If the anomaly score was 82, would you immediately raise a P2 incident? Why or why not? - What additional information would you want before escalating?		
? Check Your Understanding 2. Explain the difference between supervised and unsupervised machine learning. Why does AIOps primarily rely on unsupervised learning for anomaly detection? 2. Your ML job returns an anomaly score of 88 for a CPU metric at 2:15 AM. A colleague says 'score above 75, raise a P2 immediately.' What is your response? What additional context would you gather first? 2. Explain the difference between a deterministic and a probabilistic AIOps alert. Give one specific example of where you would use each in your operational environment at Microland.		

Capability	What It Does	Practical Value for Your Role
2. Name two practical applications of Generative AI in AIOps operations, and describe one data governance risk that must be managed before enabling Gen AI on operational log data.		

Day 3: AIOps Use Cases, Incident Management and Organisational Mindset

Duration: 4 Hours | Format: Lecture + Scenario Walkthrough + Hands-On Lab

Blueprint Domains: [Domain 5 — AIOps and Operations Metrics](#) | [Domain 6 — AIOps Use Cases and Organisational Mindset](#)

Goal: See AIOps applied to concrete IT operations scenarios you recognise from daily work. Learn how to measure AIOps impact using MTTD, MTTR, and DORA metrics. Understand the cultural and organisational changes required for AIOps to succeed — which is where most real-world AIOps programmes fail.

Time	Module	Key Topics
0:00-0:50	Module 4.1	Core AIOps use cases: event correlation, RCA, predictive maintenance, security operations, ITSM integration
0:50-1:25	Module 4.2	Measuring AIOps impact: MTTD, MTTR, alert noise reduction, DORA metrics, building the business case
1:25-2:00	Module 4.3	Organisational mindset: resistance patterns, new roles, change management, ethics and ML bias
2:00-3:00	Lab	day4_incident_simulator.ipynb: Log Rate Analysis + alert-to-Case automation
3:00-3:35	Scenario walkthrough	VMware host failure cascade: 250 alerts to 1 incident, live narration
3:35-4:00	Wrap-up	Q&A; assign Day 4 capstone pre-read scenario; feedback form

Module 4.1 — Core AIOps Use Cases

Use Case 1: Intelligent Event Correlation and Incident Compression

This is the single most impactful AIOps capability for operations teams. Consider a real scenario: a VMware ESXi host fails. Within 60 seconds, 50 VMs report ping timeouts, 50 backup jobs fail, 50 monitoring agents disconnect, and 12 application teams report service unavailability. Your monitoring system generates 250+ individual alerts and 250+ auto-raised tickets. An inexperienced L1 engineer working individual VM tickets may spend 2+ hours without ever identifying the underlying host failure.

With AIOps correlation, the ML observes: 250 alerts arrived in a 10-second window, all from systems sharing one ESXi host. It creates 1 incident record: root cause — ESXi host HA-HOST-04, severity Critical, assigned to the virtualisation team, all 50 affected VMs listed.

- Temporal correlation: 250 alerts in 10 seconds are statistically one event, not 250
- Topology correlation: all 50 affected VMs share one host — the host is the common dependency
- Historical correlation: this pattern was seen 3 months ago and resolved by addressing the host

□ Key Takeaway

Event correlation does not just reduce alert volume — it fundamentally changes the nature of the work. Instead of 250 identical tickets, one engineer works one high-quality incident with full context already assembled by the ML.

Industry benchmarks show AIOps-enabled teams reduce MTTD from an average of 4+ hours to under 15 minutes for infrastructure-level failures. That is the difference between a user-reported outage and an operations-detected, pre-user-impact response.

Use Case 2: Predictive Maintenance and Capacity Management

- Disk exhaustion: ML forecasts disk fill date weeks in advance — change request raised before the crash
- Memory leaks: progressive RAM increase detected by trend analysis before service degradation occurs
- Network saturation: bandwidth forecasting for WAN links ahead of quarter-end or product launch events
- Hardware failure prediction: correlation of early-warning signals — increased error rates, temperature trends, SMART disk metrics — before hardware fails

Use Case 3: Security Operations — AIOps and SecOps

- Anomalous user behaviour: account logging in at 3 AM from a new country scores 97 — human review triggered before any damage occurs
- Lateral movement detection: unusual sequence of logon events across multiple servers flagged as a correlated security incident
- Brute force detection: 1,000 failed logins from one IP in 5 minutes — ML detects the count anomaly before a static threshold would have triggered

Use Case 4: Intelligent ITSM Ticketing

AIOps platforms integrate with ServiceNow and Jira to auto-create enriched incident records. Instead of a bare 'CPU alert on SERVER-42', the auto-created ticket contains: the ML-identified probable root cause and confidence score, all correlated events from the same time window, historical context showing how this pattern was previously resolved, and a suggested assignment group based on the failure type.

Module 4.2 — Measuring AIOps Impact

Primary Operations Metrics

Metric	Definition	Typical AIOps Impact
MTTD (Mean Time to Detect)	Average time from when an issue starts to when the operations team identifies it	Without AIOps: industry average 4+ hours. With AIOps Level 2: typically under 15 minutes for infrastructure failures
MTTR (Mean Time to Resolve)	Average time from detection to full service restoration	AIOps reduces MTTR by providing pre-assembled context, suggested runbooks, and auto-remediation for known issue types
Alert Noise Reduction	Percentage reduction in alert volume after intelligent correlation	Typical range: 60-90% reduction with event correlation enabled

Metric	Definition	Typical AIOps Impact
False Positive Rate	Percentage of alerts raised that required no engineering action	Key quality metric for ML job tuning — a high false positive rate erodes engineer trust in the platform
Incident Volume	Total incidents raised per month or quarter	AIOps Level 2-3 teams see 30-50% reduction through proactive issue prevention before users are impacted

DORA Metrics — Linking AIOps to Business Outcomes

DORA (DevOps Research and Assessment) metrics are the industry standard for measuring operational performance. AIOps directly improves several:

DORA Metric	What It Measures	How AIOps Improves It
Deployment Frequency	How often code is deployed to production	Faster incident detection and resolution enables more confident, higher-frequency deployments
Change Failure Rate	Percentage of deployments causing a production incident	AIOps correlates deployments with post-deployment anomalies, enabling rapid rollback decisions within minutes
Failed Deployment Recovery Time	Time to restore service after a failed deployment	ML-driven RCA and automated remediation significantly reduce time to recovery
Rework Rate	Percentage of unplanned deployments caused by production incidents	Predictive AIOps prevents the incidents that drive reactive rework deployments

Building the Business Case — A Simple Framework

If asked to justify AIOps investment, use this calculation framework:

- Cost of alert fatigue: (alerts per day) x (% requiring action) x (avg triage time in minutes) x (engineer hourly cost) x (working days per year)
- Value of MTTD improvement: (P1 incidents per year) x (avg revenue or SLA penalty per hour of downtime) x (expected MTTD reduction in hours)
- Value of incident prevention: (outages from predictable causes — disk full, memory leak) x (avg resolution cost) x (forecast prevention rate: 40-60%)
- Compare against tool cost: Elastic Cloud starts at approximately USD 95/month per deployment; Grafana Cloud has a generous free tier

Module 4.3 — Organisational Mindset and Change Management

Technology is only 30% of an AIOps transformation. The remaining 70% is people and process. AIOps programmes consistently fail not because the ML does not work, but because the organisation is not ready to trust or act on its outputs.

Common Resistance Patterns and How to Address Them

Resistance Pattern	Why It Happens	How to Respond
'The AI is wrong' (after one false positive)	Engineers lose trust after a single incorrect ML recommendation and dismiss the system entirely	Show the true positive rate alongside the false positive rate. One false positive in 100 detections is still 99% noise reduction. Build trust incrementally with lower-stakes use cases first.
'It will replace my job'	Fear of automation making human engineers redundant	AI Ops reduces toil — alert triage, ticket logging, manual checks. It elevates engineers to higher-value diagnostic and strategic work. Document what the team does with reclaimed time to make the benefit visible.
'Our environment is too complex'	Belief that ML models cannot handle a specific environment's complexity	AI Ops learns YOUR environment specifically. The models are not pre-trained on generic data — they train on your actual operational data over time.
'We need more data first'	Reluctance to start until the data quality is 'perfect'	Start collecting now. You need 2-4 weeks of data before ML is reliable. Waiting for perfection delays value by months. Start with one data source and expand.

New Roles Emerging in AIOps Environments

- AIOps Engineer: Configures ML jobs, tunes anomaly detection thresholds, manages data ingestion pipelines, integrates AIOps platform with ITSM tools
- Data Quality Owner: Ensures operational data is consistently normalised, timestamped correctly, and complete before ML consumes it
- Automation Steward: Reviews and approves automated remediation playbooks; defines Human-in-the-Loop guardrails for each automation type
- Observability Lead: Designs the data collection architecture; ensures the Five V's — particularly Veracity — are actively managed

Ethics and ML Bias in AIOps

ML models in AIOps learn from historical data — which means they can encode historical patterns as permanent definitions of 'normal'. This creates specific operational risks:

- Maintenance window bias: if your environment has a recurring weekly maintenance window, the ML learns it as expected and may suppress genuine failures during that window
- New infrastructure blindness: when new servers or services are added, the ML has no baseline — expect higher false positive rates for 2-4 weeks as it learns
- Seasonal drift: models trained on summer workload patterns may misfire when winter demand patterns arrive. Schedule quarterly model retraining cycles.
- Alert suppression bias: if engineers consistently dismiss alerts from a source as noise, the ML learns to down-score that source — genuine issues from it may then be missed

□ Hands-On Lab: Day 3 — Intelligent Incident Investigation

Duration: 60 minutes | Platform: Elastic Cloud + Google Colab

Notebook: day4_incident_simulator.ipynb

Objective: Use Log Rate Analysis to identify the root cause signal in a simulated incident, then create an alert rule that automatically opens a Case (simulated ITSM ticket).

Part A — Generate and Upload Incident Data (15 min):

1. Open day4_incident_simulator.ipynb in Google Colab.
2. Paste your Cloud ID and API Key into the configuration cell.
3. Run all cells. The notebook generates a 4-hour log stream with a burst of 500 failed authentication events injected at the 2-hour mark — simulating a brute-force attack.
4. Verify data in Kibana under index: aiops-lab-day4.

Part B — Log Rate Analysis (Root Cause Investigation) (20 min):

1. Kibana > Machine Learning > Log Rate Analysis.
2. Index: aiops-lab-day4 | Set time range to cover the full 4-hour window.
3. Click Run Analysis.
4. Observe which log terms are statistically overrepresented in the anomalous window compared to the baseline window.
5. Record: which ECS field and value did the analysis flag as the primary change?
6. Question: without Log Rate Analysis, how long would it take to find this in 4,000 events?

Part C — Create an Alert that Opens a Case (25 min):

1. Kibana > Stack Management > Rules and Connectors > Create Rule.
2. Rule type: Elasticsearch query.
3. Condition: count of event.action: authentication_failure exceeds 50 in a 5-minute window.
4. Action: Create Case with title, description (include source IP and count), severity: High.
5. Test the rule by re-running the spike injection cell in the Colab notebook.
6. Navigate to Cases and verify the auto-created incident record.

Reflection (in the last 5 minutes):

Compare the auto-created Case record with what an L1 engineer would write manually. What information is pre-populated? What is missing that a human would add? What does this tell you about the current limits of AIOps automation?

? Check Your Understanding

3. A VMware host failure causes 80 VMs to report as unreachable. Without AIOps, how many individual alert notifications might the L1 team receive? With AIOps intelligent correlation, what is the ideal outcome in terms of incident records and information quality?
3. Your team's current MTTD is 4.5 hours. After deploying AIOps Level 1 capabilities, it drops to 45 minutes. Calculate the percentage improvement. Beyond the number, what does this change mean operationally for your customers?

3. Your most experienced engineer says 'I do not trust the AI — it was wrong last week.' Describe the specific approach you would take to rebuild their trust in the ML outputs over the next 30 days.
3. Explain what ML bias means in an AIOps context. Describe one specific example from IT operations where historical data in your environment could lead to a biased model that creates a blind spot.

Day 4: Evaluating Impact, Implementing AIOps and Capstone

Duration: 4 Hours | Format: Lecture + Capstone Lab + Final Assessment

Blueprint Domains: Domain 7 — Evaluating AIOps Impact | Domain 8 — Implementing AIOps in the Organisation

Goal: Apply the full Observe-Think-Act loop in a complete end-to-end capstone scenario. Understand how to evaluate AIOps effectiveness post-deployment, navigate the governance and regulatory requirements for implementation, and chart your personal certification and career path forward.

Time	Module	Key Topics
0:00-0:35	Module 5.1	Evaluating AIOps effectiveness: tracking ML accuracy, false positive management, model governance, feedback loops
0:35-1:05	Module 5.2	Implementing AIOps: paths to adoption, data governance, regulatory considerations, implementation pitfalls
1:05-2:35	Capstone Lab	day5_memory_leak.ipynb: full Observe-Think-Act loop — memory leak detection, ML alert, webhook notification
2:35-3:00	Assessment	Final quiz: 20 multiple-choice questions covering all 8 PeopleCert blueprint domains
3:00-3:40	Career Roadmap	Certification pathways, recommended next steps, free resources, continuing education plan
3:40-4:00	Wrap-up	Programme debrief, feedback, certificates of completion, Q&A

Module 5.1 — Evaluating AIOps Effectiveness

Deploying AIOps is not the end — it is the beginning. The platform needs ongoing measurement, tuning, and governance to deliver sustained value. Most AIOps programmes that fail do so not because the technology does not work, but because the organisation stopped measuring and tuning after the initial go-live.

How to Track and Improve ML Accuracy Over Time

- Track False Positive Rate weekly: if more than 20% of ML alerts require no action, the model needs retraining or threshold adjustment for that metric
- Implement a structured feedback loop: require engineers to mark every alert as 'Genuine Issue' or 'False Positive' — this feedback is the most valuable input for improving model accuracy
- Schedule model retraining: at minimum quarterly, and immediately after any major infrastructure change — new datacentre onboarding, cloud migration, major application deployment
- Monitor model freshness: most platforms display the last-trained date for each model. A model trained on last year's data will not reflect current workload patterns.

- Track anomaly score distribution: if 80% of your anomaly scores cluster between 50-60 with few high-confidence detections, your model may need more training data or feature engineering

Governance Requirements for Production AIOps

- Explainability: every alert must show which signals contributed to it and their relative weight. 'The ML said so' is not an acceptable explanation in a post-incident review or regulatory audit.
- Auditability: every automated action must produce a permanent, immutable log entry recording what triggered it, what was executed, who or what approved it, and what the outcome was
- Change control: changes to ML job configurations, detection functions, or automation playbooks must follow a defined change management process — they are production changes
- Data lineage: be able to trace every ML output back to its source data — essential for identifying whether a model failure was caused by a data quality problem upstream

□ Key Takeaway

Treat your ML models like production applications — they need monitoring, maintenance, testing, and governance just like your servers do. A model that has not been retrained after a major infrastructure change is a liability, not an asset.

The first 90 days after go-live are the most critical for AIOps adoption. A high false positive rate during this window causes engineers to lose trust permanently. Prioritise reducing false positives over expanding coverage in the first three months.

Module 5.2 — Implementing AIOps in the Organisation

Recommended Implementation Paths

Path	Approach	Best For
Start with observability	Deploy centralised logging and metrics first. Add ML once data quality is established and the team understands the data.	Organisations with fragmented, siloed monitoring today — the most common starting position
Augment existing ITSM	Integrate AIOps intelligence into ServiceNow or Jira without replacing monitoring tools. Enrich existing tickets with ML context.	Organisations where ITSM is the operational centre of gravity and change management buy-in must come from service management teams
Pilot on one service	Select a high-volume, high-noise service (e.g. Windows security events) and prove value on that service before wider rollout.	Organisations that need to demonstrate ROI to leadership before committing to broader investment
Platform consolidation	Replace fragmented legacy monitoring tools with a unified AIOps platform across all teams.	Organisations with mature data strategy, executive sponsorship, and high tolerance for disruption during transition

Data Governance and Regulatory Considerations

For IT service organisations like Microland serving clients in regulated industries — financial services, healthcare, and government — these governance requirements apply before implementing AIOps:

- Data residency: verify that your Elastic Cloud deployment region complies with client data sovereignty requirements (e.g. data from Indian banking clients must remain within India)
- Personal data in logs: operational logs often contain usernames, IP addresses, session IDs, and browsing patterns that may qualify as personal data under GDPR or India's Digital Personal Data Protection Act. Implement log masking or field anonymisation before ingesting into AIOps platforms.
- Explainability for regulated decisions: in some regulated contexts, automated decisions based on ML outputs may need to be explained to auditors. Verify your AIOps platform can produce explainability reports.
- Retention policies: operational data fed to ML models must have defined retention periods. Indefinitely retaining security event logs creates both compliance risk and storage cost.
- Model transparency: document which ML model version produced each alert in production. If a model version is found to be biased or inaccurate, you must be able to identify all alerts it produced.

Avoiding Common Implementation Pitfalls

- Pitfall 1 — Boiling the ocean: trying to connect all data sources before proving value on any single source. Start with the highest-volume, highest-pain data source and expand from there.
- Pitfall 2 — Skipping data quality: enabling ML jobs before data normalisation is complete. The model trains on bad data and produces bad results, destroying trust before the platform has a chance to prove value.
- Pitfall 3 — No human-in-the-loop for automation: enabling automated remediation without a defined approval workflow. A misconfigured automation that restarts the wrong service at 2 AM causes more damage than the original alert.
- Pitfall 4 — No success metrics: deploying AIOps without baseline measurements of MTTD, MTTR, and alert volume. Without a before-and-after comparison, you cannot demonstrate or quantify the value.

□ Hands-On Lab: Day 4 Capstone — The Full AIOps Loop: Memory Leak Scenario

Duration: 90 minutes | Platform: Elastic Cloud + Google Colab + webhook.site

Notebook: day5_memory_leak.ipynb

Scenario: A Windows application service on server 'microland-app-03' is experiencing a slow

memory leak. RAM consumption has been rising 2% per hour for the past 24 hours.

The service will crash in approximately 18 hours if no action is taken.

Your task: implement the complete Observe-Think-Act loop.

OBSERVE — Step 1: Generate and ingest the data

1. Open day5_memory_leak.ipynb in Google Colab.
2. Run all cells. The notebook generates 72 hours of RAM utilisation metrics:

- Hours 0-48: stable usage at ~55% (the baseline the ML will learn)
Hours 48-72: progressive increase at +2% per hour (the simulated leak)
3. Verify data in Kibana under index: aiops-capstone (should show 8,640 documents).

THINK — Step 2: Create the anomaly detection job

1. Kibana > Machine Learning > Create Job > Single Metric.
2. Index: aiops-capstone | Function: high_mean | Field: system.memory.used.pct
3. Job ID: capstone-memory-leak | Bucket span: 1h
4. Run the job across all 72 hours of data.
5. Open Anomaly Explorer. Verify the job detects the anomaly in the rising portion.
6. Record: the exact timestamp and anomaly score of the first flagged bucket.

THINK — Step 3: Create a 24-hour forecast

1. Single Metric Viewer > Forecast > Duration: 24 hours.
2. At what time does the forecast predict memory will cross 90%?
3. How does the ML's predicted crash time compare to the 18-hour scenario estimate?

ACT — Step 4: Create an alert with webhook notification

1. Go to webhook.site in a new browser tab. Copy your unique personal webhook URL.
2. Kibana > Stack Management > Rules and Connectors > Create Connector > Webhook.
URL: your webhook.site URL | Method: POST | Content-Type: application/json
3. Create a Rule: fire when ML anomaly score exceeds 75 on job capstone-memory-leak.
4. Action payload (paste this JSON into the request body field):

```
{ "server": "microland-app-03", "issue": "Memory leak detected",  
  "anomaly_score": "{{context.score}}",  
  "recommendation": "Investigate and restart the application service" }
```
5. Trigger the alert by adjusting the time window or re-running the job.
6. Verify the POST payload arrived at webhook.site with the correct fields.

Document Your Work (final 10 min):

Write a 10-line shift handover note as if passing this incident to the next team.
Include: what was detected, when it was first flagged, the ML anomaly score, the forecast impact time, what automated action was triggered, and your recommended next action for the receiving team.

Expected Outcome: A complete, documented Observe-Think-Act loop.

Data ingested > ML anomaly detected > forecast showing time to impact > automated alert fired.

Module 5.3 — Your AIOps Career Roadmap

This programme is your foundation. The AIOps and observability space is growing rapidly, and the skills you have developed over these four days are in high demand across IT service organisations. Here is a structured path forward:

Priority	Certification	Provider	Cost	Timeline
Immediate	AIOps Foundation SM	PeopleCert / DevOps Institute	~USD 400	Exam within 4 weeks of course
3 months	Elastic Certified Observability Engineer	Elastic	~USD 400	After 3 months hands-on practice
6 months	AWS Certified AI Practitioner	Amazon Web Services	USD 100	Cloud AIOps context
6 months	AI-900: Azure AI Fundamentals	Microsoft	USD 165	For Azure-heavy client environments
Ongoing	AIOps Fundamentals (self-paced)	Great Learning Academy	Free	Self-paced video refresher
Ongoing	DevOps and AI on AWS (audit)	Coursera	Free (audit)	Anytime — deepen cloud context

❏ Important Note

The PeopleCert AIOps FoundationSM exam: 40 multiple-choice questions, 60-minute duration, open-book permitted, passing score 65% (26/40 correct). All 8 domains covered in this 4-day programme appear in the exam.

Recommended immediate action: keep your Elastic Cloud trial active for the remaining days and use the built-in sample datasets (ecommerce, flights, logs) to create additional ML jobs and build your practical confidence before the trial expires.

? Check Your Understanding

- After deploying AIOps, your team's false positive rate is 38% in the first month. A senior manager interprets this as a technology failure. What is your response? What three specific actions would you take to reduce the false positive rate over the next 30 days?
- A Microland client in the banking sector wants to implement AIOps on their core banking infrastructure. Identify three data governance actions that must be completed before operational logs can be ingested into the AIOps platform.
- Using the alert fatigue cost framework: your team of 10 engineers receives 2,500 alerts per day. 25% require action. Average triage time is 10 minutes. Blended engineer cost is INR 1,500/hour. AIOps reduces alert volume by 75%. Calculate: (a) current annual cost of alert triage, (b) cost after AIOps, (c) annual saving.
- List the 8 PeopleCert AIOps Foundation blueprint domains and briefly describe what each covers. (Use your notes from all four days — this mirrors the structure of the actual certification exam.)

Appendix: Quick Reference

A. Complete Glossary of AIOps Terms

Term	Definition
AIOps	Artificial Intelligence for IT Operations — using big data, ML, and automation to enhance IT operations. Coined by Gartner in 2016 as 'Algorithmic IT Operations'.
Alert Fatigue	Operational state where engineers receive so many monitoring alerts that they become desensitised, increasing the risk of missing genuine critical events.
Anomaly Score	A 0-100 score produced by ML indicating how statistically unusual a data point is relative to the learned baseline. 0 = expected; 100 = highly anomalous.
Baseline	An ML-learned statistical model of normal behaviour for a metric or system over time. The reference point against which anomalies are measured.
Deterministic	Rule-based approach where the same input always produces the same output. Fully transparent, auditable, and reproducible. Used for known, defined conditions.
DevOps	Culture and engineering practices that unite software development and IT operations to accelerate software delivery and improve reliability.
DORA Metrics	DevOps Research and Assessment metrics: Deployment Frequency, Lead Time for Changes, Change Failure Rate, Failed Deployment Recovery Time, and Rework Rate.
ECS (Elastic Common Schema)	A standardised field naming convention that normalises operational data from diverse sources into a unified format so ML can analyse them as a single dataset.
Five V's of Big Data	Volume, Velocity, Variety, Veracity, Value — the five dimensions that characterise big data and explain why IT operational data requires specialised AI-powered platforms.
ITOA (IT Operations Analytics)	The predecessor to AIOps. Used big data collection for IT analysis but relied on static, human-authored rules applied to historical data — unable to adapt dynamically.
MLOps	Machine Learning Operations — the engineering practice of managing ML model lifecycles in production: training, deployment, monitoring, and retraining.
MTTD	Mean Time to Detect — the average elapsed time from when an incident begins to when it is identified by the operations team.
MTTR	Mean Time to Resolve — the average elapsed time from incident detection to full service restoration and closure.
Probabilistic	ML-based approach that outputs a probability or confidence score rather than a binary yes/no decision. The primary mode for AIOps anomaly detection.

Term	Definition
Self-Healing	Automated remediation triggered by AIOps without human intervention. Should be limited to well-understood, low-risk, and fully reversible actions.
SRE (Site Reliability Engineering)	Engineering discipline applying software development practices to IT operations. Uses error budgets and Service Level Objectives (SLOs) to manage system reliability.
Unsupervised Learning	ML approach where the algorithm learns statistical patterns from unlabelled data. The primary ML type used in AIOps anomaly detection.
Veracity	The accuracy and trustworthiness of data. The most commonly neglected of the Five V's — poor data quality degrades ML model accuracy more than algorithm limitations.
Webhook	An HTTP POST callback that sends a structured JSON payload to an external URL when a defined condition is met — the most common mechanism for AIOps automation triggers.

B. The 8 PeopleCert AIOps FoundationSM Blueprint Domains

#	Domain	What It Covers (and Which Day Covers It)
1	AIOps Foundation	History and evolution from ITOA; Gartner's definition; current stages of AIOps systems. (Day 1)
2	AIOps in the Organisation	Integration with DevOps, SRE, and security operations; organisational frameworks; system complexity. (Day 1)
3	Core Technologies: Big Data	The Five V's of big data; data sources and types; characteristics of operational data at scale. (Day 1)
4	Core Technologies: ML and Gen AI	Supervised vs unsupervised learning; deterministic vs probabilistic; Generative AI in AIOps; MLOps relationship. (Day 2)
5	AIOps and Operations Metrics	MTTD, MTTR, alert noise reduction, DORA metrics; tracking incident impact; improving ML accuracy. (Day 3)
6	AIOps Use Cases and Org Mindset	Real-world use cases; cultural resistance patterns; new roles; organisational change management; ethics and bias. (Day 3)
7	Evaluating AIOps Impact	Measuring effectiveness post-deployment; quantifying ROI; governance; feedback loops and model retraining. (Day 4)
8	Implementing AIOps	Paths to adoption; data quality requirements; regulatory and privacy considerations; avoiding implementation pitfalls. (Day 4)

C. Lab Notebook Index

Notebook	Content	Used On
day2_log_generator.ipynb	Generates 1,000 Windows-style ECS-aligned log events with 20	Day 1 Lab

Notebook	Content	Used On
	injected brute-force anomalies; uploads to Elastic Cloud index aiops-lab-day2	
day3_spike_simulator.ipynb	Generates 720 CPU metric data points (6 hours at 30s intervals) with a 30-minute 3x spike injected at hour 3; uploads to aiops-lab-day3	Day 2 Lab
day4_incident_simulator.ipynb	Generates a 4-hour log stream with 500 failed authentication events injected at the 2-hour mark; uploads to aiops-lab-day4	Day 3 Lab
day5_memory_leak.ipynb	Generates 72 hours of RAM metrics (48h stable baseline at 55%, then 24h progressive +2%/hr leak); uploads to aiops-capstone	Day 4 Capstone

❏ Important Note

All four notebooks are shared by the instructor via Google Drive at the start of Day 1. Each notebook is fully self-contained with step-by-step instructions. No Python knowledge is required to run them.

D. Recommended Free Resources

- Elastic documentation and tutorials: elastic.co/guide and elastic.co/learn
- Elastic ML documentation: elastic.co/guide/en/machine-learning/current/index.html
- Grafana Labs learning portal: grafana.com/tutorials
- Great Learning AIOps Fundamentals free course: mygreatlearning.com/academy
- DevOps Institute AIOps Foundation overview: devopsinstitute.com/certifications/aiops-foundation
- PeopleCert AIOps Foundation blueprint download: peoplecert.org — search 'AIOps Foundation'
- AWS DevOps and AI on Coursera (free to audit): coursera.org
- Prometheus getting started guide: prometheus.io/docs/prometheus/latest/getting_started
- OpenTelemetry documentation and getting started: opentelemetry.io/docs